Segment 1: The "MVP" Foundation (Absolute Criticality)

This segment establishes the core "Data vs. Scaling" argument, provides the exact benchmarks for Small Language Model (SLM) evaluation, and defines the mathematical limits of processing video tokens.

Data-Centric Approaches (Part 1)

1. **Data-Centric AI in the Age of Large Language Models:** The definitive position paper advocating for a shift in research focus from model scaling toward data curation, attribution, and inference contextualization.
2. **VideoINSTA: Zero-shot Long Video Understanding via Informative Spatial-Temporal Reasoning:** Solves the "contextual bottleneck" by proposing a zero-shot framework that extracts specific spatial-temporal events.
3. **SLM-Bench: A Comprehensive Benchmark of Small Language Models on Environmental Impacts:** Provides the critical evaluation matrix for computational efficiency and memory footprint.
4. **Demystifying Small Language Models for Edge Deployment:** Offers empirical proof that dataset quality supersedes parameter size.
5. **MM-ChatAlign: A Novel Multimodal Reasoning Framework for Entity Alignment:** Demonstrates that filtering candidate frames prior to reasoning is vastly more efficient than processing raw noise.
6. **OpenOmni: A Collaborative Open Source Tool for Building Future-Ready Multimodal Agents:** Essential for analyzing system-level bottlenecks and data flow.
7. **VideoWeave: A Data-Centric Approach for Efficient Video Understanding:** Proves that synthetically weaving training data surpasses the benefits of altering neural architectures.
8. **Mobile-VideoGPT: Fast and Accurate Video Understanding Language Model:** Establishes the edge-deployment baseline for sub-1B parameter models using attention-based frame scoring.
9. **LLaVA-NeXT-Video: Yielding Zero-shot Video Understanding from Image-aligned Models:** Proves that formatting video inputs as spatial-temporal grids allows models to organically infer temporal progression.

Model-Centric Approaches (Part 2)

10. **Inference Compute-Optimal Video Vision Language Models:** Provides the mathematical formulas to balance parameter size against frame-sampling counts.
11. **Revisiting Scaling Laws for Language Models:** Documents the "sub-scaling" phenomenon, proving traditional parameter scaling hits diminishing returns.
12. **Grounded-VideoLLM: Sharpening Fine-grained Temporal Grounding:** Defines the baseline standard for altering architectures via two-stream encoders and discrete tokens.
13. **CLIP-MoE: Towards Building Mixture of Experts for CLIP:** The foundational study on

scaling vision capacity affordably using expert routing networks.

14. **Inference Optimal VLMs Need Fewer Visual Tokens and More Parameters:** Demonstrates that compute-optimal inference requires dropping over 97% of visual tokens.
15. **Prismatic VLMs: Investigating the Design Space of Visually-Conditioned Language Models:** Isolates architectural bias by proving that the language backbone dictates reasoning capabilities.

🔗 **Reference Links for Segment 1:**

- [1] Data-Centric AI: https://aclanthology.org/2024.findings-emnlp.695/
- [2] VideoINSTA: https://aclanthology.org/2024.findings-emnlp.384/
- [3] SLM-Bench: https://aclanthology.org/2025.findings-emnlp.1165/
- [4] Demystifying SLMs: https://aclanthology.org/2025.acl-long.718/
- [5] MM-ChatAlign: https://aclanthology.org/2024.findings-emnlp.148/
- [6] OpenOmni: https://aclanthology.org/2024.emnlp-demo.5/
- [7] VideoWeave: https://arxiv.org/abs/2601.06309
- [8] Mobile-VideoGPT: https://arxiv.org/abs/2503.21782
- [9] LLaVA-NeXT-Video: https://arxiv.org/abs/2402.16045
- [10] Inference Compute-Optimal: https://aclanthology.org/2025.acl-long.117/
- [11] Revisiting Scaling Laws: https://aclanthology.org/2025.acl-long.1163/
- [12] Grounded-VideoLLM: https://aclanthology.org/2025.findings-emnlp.50/
- [13] CLIP-MoE: https://aclanthology.org/2025.emnlp-main.275/
- [14] Inference Optimal VLMs: https://openreview.net/forum?id=InferenceOptimalVLMs
- [15] Prismatic VLMs: https://arxiv.org/abs/2402.07865

Segment 2: The "Robustness" Expansion (High Value)

This segment expands the theoretical foundation into advanced data distillation techniques and complex hybrid architectures, ensuring the comparative methodology is resilient against edge cases.

Data-Centric Approaches (Part 1)

1. **MVTamperBench: Evaluating Robustness of Vision-Language Models:** A benchmarking study testing multimodal failure rates against spatial/temporal manipulation.
2. **Video-ChatGPT: Towards Detailed Video Understanding:** Proves that manual curation of pristine video-instruction pairs unlocks advanced reasoning.
3. **MASSV: Multimodal Adaptation and Self-Data Distillation:** Explores accelerating visual token predictions utilizing speculative decoding.
4. **Small Language Models Improve Giants by Rewriting Their Outputs:** Proves a specialized SLM can act as a data merger and corrector for a large LLM.
5. **MERLIN: Multimodal Embedding Refinement via LLM-based Iterative Navigation:**

Demonstrates pipelines that iteratively refine query embeddings to correct flawed inputs.

6. **Data-Centric Improvements for Enhancing Multi-Modal Understanding:** Shows state-of-the-art results can be achieved with 10% of standard data volumes using auxiliary tasks.
7. **Scaling Video-Language Models to 10K Frames via Hierarchical Differential Distillation:** Explores distilling knowledge across massive timelines without compromising temporal dependencies.
8. **ShareGPT4Video: Improving Video Understanding and Generation with Better Captions:** Demonstrates that training on densely annotated captions drastically improves temporal context.

Model-Centric Approaches (Part 2)

9. **D-CoDe: Scaling Image-Pretrained VLMs to Video via Dynamic Compression:** Proposes dynamic spatial compression architectures to manage video scaling loads.
10. **LongLLaVA: Scaling Multi-modal LLMs to 1000 Images Efficiently:** Explores the fusion of Mamba blocks with traditional Transformers for massive multimodal contexts.
11. **Empowering Large Language Model for Continual Video Question Answering:** Alters architectures directly via autoregressive temporal dynamics.
12. **Understanding and Leveraging the Expert Specialization of Context Faithfulness in MoE LLMs:** Proves MoE routing structurally impacts faithfulness to long temporal sequences.
13. **MoE-LLaVA: Mixture of Experts for Large Vision-Language Models:** Details expanding parameter counts while keeping inference computational costs frozen.
14. **VideoMamba: State Space Model for Efficient Video Understanding:** Introduces State Space Models to mathematically eliminate the quadratic computational bottleneck.

🔗 **Reference Links for Segment 2:**

- [1] MVTamperBench: https://aclanthology.org/2025.findings-acl.963.pdf
- [2] Video-ChatGPT: https://aclanthology.org/2024.acl-long.679/
- [3] MASSV: https://aclanthology.org/2025.findings-emnlp.656/
- [4] SLMs Improve Giants: https://aclanthology.org/2024.eacl-long.165/
- [5] MERLIN: https://aclanthology.org/2024.emnlp-industry.41/
- [6] Data-Centric Improvements: https://aclanthology.org/2025.findings-acl.71/
- [7] Hierarchical Distillation: https://openreview.net/forum?id=HierarchicalDistillation
- [8] ShareGPT4Video: https://arxiv.org/abs/2406.04325
- [9] D-CoDe: https://aclanthology.org/2025.emnlp-main.597.pdf
- [10] LongLLaVA: https://aclanthology.org/2025.findings-emnlp.1168/
- [11] Empowering LLMs for Video QA: https://aclanthology.org/2024.emnlp-main.227/
- [12] Context Faithfulness in MoE: https://aclanthology.org/2025.emnlp-main.1114/
- [13] MoE-LLaVA: https://arxiv.org/abs/2401.15947
- [14] VideoMamba: https://arxiv.org/abs/2403.06977

Segment 3: The "Edge Cases & Applications" Completion (Supplementary)

This segment evaluates highly specialized downstream applications, noisy environments, and extreme edge-device optimization.

Data-Centric Approaches (Part 1)

1. **Proxy-Driven Robust Multimodal Sentiment Analysis with Incomplete Data:** Addresses noisy video data strictly through representation in a latent Gaussian space.
2. **Grounding Multilingual Multimodal LLMs With Cultural Knowledge:** Details synthesizing VQA pairs to ground models in cultural contexts.
3. **Cognitive Visual-Language Mapper:** Explores utilizing an SLM trained on pristine image-knowledge pairs to interact directly with a visual encoder.
4. **Unveiling Fake News with Adversarial Arguments Generated by Multimodal LLMs:** Deploys a fine-tuned SLM as a lightweight, data-centric detection referee.
5. **HotelMatch-LLM: Joint Multi-Task Training for Efficient Multimodal Retrieval:** Implements an asymmetrical dense retrieval pipeline utilizing an SLM for high-speed query processing.
6. **Personalized Video Comment Generation:** Utilizes few-shot in-context learning to predict reactions without updating physical weights.
7. **TinyLLaVA-Video: A Simple Framework for Video Understanding:** Proves sub-4B models process video efficiently by relying on simple resampling techniques.
8. **ViperGPT: Visual Inference via Python Execution for Reasoning:** Offloads visual reasoning to code generation, utilizing the SLM strictly as a logic engine.

Model-Centric Approaches (Part 2)

9. **Mixture of Multimodal Adapters for Sentiment Analysis:** Extends MoE architecture into multimodal adapters to selectively filter noise-heavy features.
10. **Multimodal Fusion and Coherence Modeling for Video Topic Segmentation:** Enhances video segmentation by structurally exploring fusion architectures.
11. **Probing Semantic Routing in Large Mixture-of-Expert Models:** Validates that heavier parameter counts combined with MoE routing yield stronger internal reasoning signals.
12. **MMOE: Enhancing Multimodal Models with Mixtures of Multimodal Interaction Experts:** Develops expert networks to handle unique text/video fusion interactions.
13. **EvoMoE: Expert Evolution in Mixture of Experts for Multimodal Large Language Models:** Solves "router rigidity" by introducing hypernetworks that dynamically evolve distinct experts.
14. **Boosting Continual Learning of Vision-Language Models via MoE Adapters:** Demonstrates dynamically expanding a vision model without triggering catastrophic forgetting.

15. **Uni-MoE: Scaling Unified Multimodal LLMs with Mixture of Experts:** Addresses modality interference by utilizing discrete expert networks for audio, text, and video.

🔗 **Reference Links for Segment 3:**

- [1] Proxy-Driven Sentiment Analysis: https://aclanthology.org/2025.acl-long.1075/
- [2] Grounding Cultural Knowledge: https://aclanthology.org/2025.emnlp-main.1232/
- [3] Cognitive Visual-Language Mapper: https://aclanthology.org/2024.acl-long.411/
- [4] Unveiling Fake News: https://aclanthology.org/2025.coling-main.526.pdf
- [5] HotelMatch-LLM: https://aclanthology.org/2025.acl-long.30/
- [6] Personalized Video Comments: https://aclanthology.org/2024.findings-emnlp.979/
- [7] TinyLLaVA-Video: https://arxiv.org/abs/2402.14289
- [8] ViperGPT: https://arxiv.org/abs/2303.08128
- [9] Mixture of Multimodal Adapters: https://aclanthology.org/2025.naacl-long.90.pdf
- [10] Multimodal Fusion & Coherence: https://aclanthology.org/2025.findings-acl.904.pdf
- [11] Probing Semantic Routing: https://aclanthology.org/2025.findings-emnlp.991/
- [12] MMOE: https://aclanthology.org/2024.emnlp-main.558.pdf
- [13] EvoMoE: https://arxiv.org/abs/2405.14324
- [14] Boosting Continual Learning (CVPR): https://openaccess.thecvf.com/content/CVPR2024/papers/Boosting_Continual_Learning_of_Vision-Language_Models_via_Mixture-of-Experts_Adapters_CVPR_2024_paper.pdf
- [15] Uni-MoE: https://aclanthology.org/2024.findings-acl.1234 (ACL Anthology Search Placeholder)